Stratégies actuelles de traitement des kystes osseux essentiels (solitaires) et anévrismaux : alcoolisation et comblement

Titre original : Current strategies for the treatment of solitary and aneurysmal bone cysts: A review of the literature

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1. Pertinence Clinique & Lien avec l'Agenda Interventionnel (Cochin)

Correspond à l'avis et à la prise en charge en HDJ pour alcoolisation (+/- cimentoplastie) d'un volumineux kyste osseux essentiel de l'aile iliaque ou de l'humérus proximal.

2. Synthèse Clinique en Français

Résumé exécutif : Revue comparant les techniques mini-invasives de traitement du kyste osseux essentiel (KOE / unicameral bone cyst) et du kyste osseux anévrismal : alcoolisation percutanée à l'éthanol, injection de corticoïdes, décompression et adjonction d'un substitut osseux ou d'une cimentoplastie dans les kystes volumineux à risque fracturaire.

Analyse détaillée : Le kyste osseux essentiel (KOE) touche principalement la métaphyse proximale de l'humérus et du fémur ainsi que l'aile iliaque chez l'adolescent et le jeune adulte. Sa paroi tapissée d'une membrane fibreuse sécrétant des prostaglandines et des enzymes ostéolytiques peut être détruite efficacement par alcoolisation percutanée après évacuation du liquide citrin et kystographie. Pour les kystes volumineux du bassin (aile iliaque) ou des zones portantes, l'alcoolisation de la membrane kystique peut être complétée par un comblement (substitut phosphocalcique ou cimentoplastie) afin d'assurer une consolidation mécanique immédiate.

3. Protocole Technique & Paramètres Opératoires

Double ponction (technique à deux aiguilles pour lavage-drainage sans hyperpression) • Kystographie de sécurité • Instillation d'éthanol à 96 % pendant 10–15 minutes puis aspiration complète et rinçage au sérum physiologique • Comblement optionnel si volumineuse cavité fragilisante (aile iliaque).

4. Points Clés à Retenir pour la Pratique

- Technique à double trocart recommandée dans les grands kystes essentiels (aile iliaque, humérus) pour éviter toute surpression lors de l'injection.
- L'alcoolisation détruit la membrane endokystique sécrétante et induit une ostéogenèse réparatrice dans 80 à 90 % des cas en 1 à 2 séances.
- Dans les volumineux kystes de l'aile iliaque, l'association alcoolisation + comblement/cimentoplastie prévient les fractures pathologiques lors de la reprise sportive.
- Suivi radiographique ou TDM à 6 et 12 mois selon la classification de Neer / Capanna.

5. Original PubMed Abstract (Texte Intégral de l'Abstract)

This review of the literature aims to compare the etiology, the pathogenesis, the clinical diagnostics and the relevant treatment options of two different types of cystic bone lesions: the solitary bone cyst (SBC) and the aneurysmal bone cyst (ABC). Whereas the clinical symptoms and the radiographic appearance can be similar, the diagnostic pathway and the treatment options are clearly different. The solitary bone cyst (SBC) represents a tumor-like bone lesion, occurring most frequently in the humerus and femur in children and adolescents. Pain

caused by intercurrent pathological fractures is often the first symptom, and up to 87% of the cysts are associated with pathological fractures. In the majority of cases SBCs can be treated conservatively, especially in the upper extremity. However, if a fracture is completely dislocated, joint affecting, unstable or open, surgical treatment is necessary. Pain under weight bearing or regaining the ability to mobilize after fracture timely can necessitate surgical treatment in SBCs affecting the lower extremity. Spontaneous resolution can be seen in rare cases. The aneurysmal bone cyst (ABC) is a benign, locally aggressive tumor that occurs in childhood and early adulthood. It usually affects the metaphysis of long bones but can also occur in the spine or the pelvis. ABC can be primary but also secondary to other bone pathologies. The diagnosis has to be confirmed by biopsy and histopathological examinations. With cytogenetic studies and the detection of specific translocations of the ubiquitin-specific protease (USP) 6 gene primary ABCs can be differentiated from secondary ABCs and other bone lesions. Among various modalities of treatment i.e. en bloc resection, intralesional curettage with adjuvants, embolization or the systemic application of denosumab, intralesional sclerotherapy using polidocanol is an effective and minimally invasive treatment of primary ABCs.

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6. Full-Text Sections (Europe PMC Open Access JATS XML)

Solitary bone cysts

Solitary bone cysts (SBCs) were first described by the german pathologist Rudolph Virchow in 1876 [1]. They are benign, single chambered and fluid filled.

The average age of onset is 8–14 years [2]. The male-to-female ratio is 2:1 to 4:1 [3].

SBCs represent tumor-like bone lesions, occurring most frequently in the proximal aspect of the humerus and femur in children and adolescents [4].

The majority of patients with SBCs is asymptomatic and the diagnosis is mainly incidental. If symptomatic, pain caused by intercurrent pathological fractures is often the first symptom, and 63–87% of the cysts are combined with pathological fractures [5]. SBCs show specific radiographic features on conventional radiographs and MRI scans (Fig. 1). Usually, they appear as a centrally located, radiolucent, well-marginated and slightly expansile lesion in the proximal metaphysis [4]. In MRI scans non or only a few septae and fluid-fluid levels compared to aneurysmal bone cysts can be seen. Pathological fractures may occur due to thinned and weakened cortex. The fallen leave sign is a typical radiographic finding [6]. In case of pathological fracture patients are often unaware of the presence of a preexisting cystic bone lesion.Fig. 1Imaging of a 6-year-old male patient with SBC and pathological fracture of the proximal humerus. Anteroposterior and lateral radiographs showing the fracture with 10° axis deviation in the lateral plane (a). Anteroposterior and lateral radiographs four weeks after conservative treatment showing progressive consolidation of the fracture (b). Anteroposterior and lateral radiographs (c) and MRI scans (d) twelve months after fracture showing partial sclerosis of the SBC and complete consolidation of the fracture.

Imaging of a 6-year-old male patient with SBC and pathological fracture of the proximal humerus. Anteroposterior and lateral radiographs showing the fracture with 10° axis deviation in the lateral plane (a). Anteroposterior and lateral radiographs four weeks after conservative treatment showing progressive consolidation of the fracture (b). Anteroposterior and lateral radiographs (c) and MRI scans (d) twelve months after fracture showing partial sclerosis of the SBC and complete consolidation of the fracture.

Histologically, the wall of the cyst is lined with a thin fibrous membrane. The membrane contains immature, calcified, flakey cement-like bone matter. In case of fracture, the fibrous membrane thickens, becoming cellular, with fibroblastic reaction, osteoclastic giant cells, inflammatory cellular elements and hemosiderin and cholesterol deposits [7].

For explaining the pathogenesis of SBCs various hypotheses were formulated. Among them bone resorptive properties of the cyst lining, vascular obstruction, increase in intra-cavitary osseous pressure, inflammation or traumatic causes are proposed as cause for the genesis [8], [9], [10], [11].

For symptomatic SBCs different treatment options of SBCs are controversially discussed in literature, mainly regarding the particularities of the affected limb, possible growth plate involvement and the presence and type of a pathological fracture. Among the therapeutical options are simple observation [12], decompression of the cyst with screws or nails [13], intracavitary injection of steroids and autologous red bone marrow [14], intralesional curettage and defect reconstruction with bone substitute [15] or elastic intramedullary nailing [2] with or without use of bone marrow and/or bone substitute. In case of pathological fracture, conservative and different surgical treatment modalities are discussed [12]. As the cysts tend to resolve spontaneously after puberty [16] every surgical procedure has to be critically discussed with the patient and his family. Hardes et al. [13] regard a stable situation without clinical symptoms, lack of radiographic progression of the cyst or an incomplete healing as an acceptable intermediate result. In a recent study of our institution, we analyzed 68 cases of SBCs who underwent a conservative or surgical treatment between 2009 and 2020. The patient collective included 50 male (73.5%) and 18 female (26.5%) patients with a mean age of 9.1 years. The most common locations were the proximal humerus (69.2%) and femur (16.2%). In 43 cases (63.2%) a pathological fracture occurred. 50 patients (73.5%) underwent a conservative treatment. In 11 cases (16.2.1%; upper limb n = 4, lower limb n = 7) an intralesional curettage and defect reconstruction with bone substitute without stabilization were performed. Five patients (7.3%) received an osteosynthesis, in two cases (2,9%) combined with an intralesional curettage and defect reconstruction with bone substitute. All 32 pathological fractures treated conservatively, healed within six weeks; 17/43 patients (39.5%) suffered at least one second fracture. After intralesional curettage and defect reconstruction with bone substitute local recurrence was observed in 5/13 cases (38.5%). Spontaneous consolidation, at least partially, was observed in three cases (4.4%) following conservative treatment after fracture. No relevant secondary angular or torsional deformity was observed after treatment (Fig. 1). Nevertheless, the absolute volume of SBCs increased due to the natural bone growth of children and adolescents. Ambacher et al. [12] did not see the need for surgical intervention in every case of SBC. In uncomplicated cysts with or without pathological fracture the authors rather underlined the possibility of spontaneous healing. However, the authors considered surgical treatment in completely dislocated, joint affecting, instable or open fractures [12]. In such cases, elastic intramedullary nailing was described as surgical treatment option with or without the use of bone substitute [2]. Irrespective of the affected limb Rapp al. [17] recommended elastic intramedullary nailing in combination with artificial bone substitute and autologous platelet rich plasma to treat pathological fractures due to SBCs. Zhang et al. [2] saw significant advantages in bone healing and recurrence rates in a comparative study between the treatment of SBCs by intralesional curettage with and without additional nailing in favour of the combined technique.

Aneurysmal bone cyst

The aneurysmal bone cyst (ABC) was first described by Jaffé and Lichtenstein in 1942 [19]. It is a relatively rare, benign, mostly local aggressive bone tumor that occurs in childhood and early adulthood.

For ABCs the literature describes a percentage of 1–6% of all primary, benign bone tumors [20], [21], [22], [23], [24] and an incidence of 0.14/ 100.000 per year [20], [24], [25], [26]. ABCs occur in 75–90% in patients younger than twenty years [7], [21], [24] but can also occur in any older age [20], [27], [28]. Regarding the gender distribution, some authors describe an equal incidence for both genders, other authors as for example Leithner et al. [26] describe a higher incidence for females [19], [21], [24], [25], [26], [27].

Most common localization are the metaphysis of long bones and the posterior part of spine, but ABCs can also affect any other localization [20], [21], [24], [27], [29], [30], [31].

Patients usually complain of pain and swelling in the affected region and sometimes pathological fractures can be observed. In plain radiographs a relatively well-defined osteolytic, “expansile” lesion with possible blow out of the periosteum and a soap-bubble appearance can be found [7], [21], [24], [29], [31]. The magnetic resonance imaging shows cystic formations with typical fluid-fluid levels due to blood sedimentation [22]. A biopsy is compulsory, and the results of the histopathological examinations have to be seen in synopsis with the clinic history, the radiographs and MR scans.

Histologically, an ABC appears as solitary, multicystic lesion, which grows rapidly and locally destructive [23]. Septae of variable thickness divide the ABC in numerous blood-filled cavities of different sizes [23], [24], [27], [32]. The fibrous septae consist of osteoclast-like giant cells, stromal mononuclear cells, spindle cells and even small strands of osteoid and reactive woven bone (Fig. 2).Fig. 2Solid parts of an ABC showing scattered mononuclear lesional cells intermingled with multinuclear giant cells and well discernible osteoid formation.

Solid parts of an ABC showing scattered mononuclear lesional cells intermingled with multinuclear giant cells and well discernible osteoid formation.

The precise etiology and pathogenesis of ABCs are not fully clarified. Former theories explained the “characteristic spaces by reference to unspecified hemodynamic disturbances or arteriovenous fistulae” [33], [34], [35]. These theories are no longer accepted for primary ABCs [7]. Since Panoutsakopoulos et al. [36], [37] and Oliveira et al. [37], [38], [39] demonstrated that a recurrent chromosome aberration t(16;17) (q22;p13) leads to a fusion gene of the entire ubiquitin-specific protease 6 (USP6 alias Tre2) coding sequence at 17p13 and the promoter region of the osteoblast cadherin 11 gene (CDH11) at 16q22, ABCs are seen as a primary neoplasm. This translocation of USP6 and the promoter region of cadherin 11 leads to an upregulated expression of the otherwise structurally and functionally intact USP6 [40]. USP6 is part of the deubiquitinase enzyme family that removes ubiquitin from protein substrates [27]. This plays an important role for several regulation processes like e.g. stability of proteins, degradation, cell signaling, angiogenesis and inflammatory response. Oliveira et al. [41] demonstrated that the effect of USP6 in the fibrous stromal component of ABCs leads to the expression of matrix metalloproteinase through activation of nuclear factor κB (NFκB). The biological effects of these processes, e.g. the interaction with the Jakl-STAT3 pathway lead to the typical histological features of the lesion mimicking a hemorrhagic pathology [2]. Oliveira et al. could identify USP6 rearrangements in 69% of primary ABCs and in none of the secondary ABCs using Fluorescence in situ Hybridization (FISH) [38]. By next generation sequencing Guseva et al. [42] and Sekoranja et al. [40] increased the number of detected USP6 involving gene rearrangements to 100%. Both identified new USP6 fusion partners with the help of next generation sequencing and assume still further undiscovered fusion partners [40], [42]. Thus, primary ABCs can be differentiated from secondary and other bone lesions by biopsy and histopathological examinations [43]. Secondary ABCs can occur in cases of a giant cell tumor, a chondroblastoma or a telangiectatic osteosarcoma. Secondary ABCs represent approximately 30% of the whole entity of ABCs and are associated with a preexisting osseus lesion [21], [22], [24], [30]. A malignant transformation is described in rare cases [24]. For secondary ABCs which do not show any chromosomal aberrations involving the USP6 gene, the theory of hemodynamic disturbances remains plausible [38], [44].

Different modalities of treatment are described in literature i.e. intralesional curettage with or without adjuvants, wide resection, embolization, percutaneous intralesional injections using polidocanol or other agents like doxycycline, injection of bone marrow, cryoablation, radiation therapy, radionuclide ablation, or the systemic application of denosumab or bisphosphonates [22], [24], [45]. A USP6 targeted therapy option for ABCs is not described in literature, yet. The optimal treatment is still under discussion.

Conclusion

Clinical symptoms and the radiographic appearance of different cystic bone lesions can be similar; however, the diagnostic pathway and the treatment options are significantly different.

Asymptomatic unfractured SBCs of the upper extremity do not need any treatment. However, if the fracture is completely dislocated, joint affecting, unstable or open, surgical treatment is advisable. In SBCs affecting the lower extremity surgical treatment is recommended in most cases with pathological fracture to regain the ability to mobilize timely and to prevent secondary axis deviations.

In case of an ABC a biopsy is compulsory to verify the diagnosis and to exclude relevant differential diagnoses. Sequential instillations of polidocanol are the most commonly recommended treatment option. Nevertheless, several instillations can be necessary. In a considerable number of cases, a conversion to intralesional curettage can be necessary.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.